

Technical Specification: Automatic Parking & Peripheral Integration

BFMC Firmware Module - STM32 Implementation

1 Hardware Interface & Sensor Fusion

The firmware integrates a heterogeneous sensor suite to handle mapping and environmental awareness.

1.1 Magnetic Encoder Odometry & Mapping

Position tracking is handled by high-resolution magnetic encoders interfaced via STM32 hardware timers in **Encoder Mode**. These provide the differential pulses $(\Delta\phi_R, \Delta\phi_L)$ used for real-time pose estimation and local mapping.

$$\Delta s = \frac{r}{2}(\Delta\phi_R + \Delta\phi_L), \quad \Delta\theta = \frac{r}{L}(\Delta\phi_R - \Delta\phi_L) \quad (1)$$

Where r is the wheel radius and L is the track width. This data allows the system to map the parking spot dimensions and maintain a local coordinate system relative to the entry point.

1.2 US-100 Ultrasonic Obstacle Detection

The **US-100** sensor provides distance data via asynchronous trigger/echo pulses.

- **Triggering:** Periodic polling at 50ms intervals.
- **Safety Logic:** If an obstacle is detected at $d < 10$ cm, a safety timer is initialized.
- **Acoustic Alert:** If $d < 10$ cm persists for $T \geq 10$ s, the **Horn** (GPIO High) is activated for a fixed duration of 5 s.

1.3 Ambient Light Sensing (LDR)

An LDR is interfaced with a 12-bit ADC. The firmware applies a low-pass filter to the raw ADC values to determine lighting conditions:

- **Thresholding:** If $ADC_{val} < \text{Threshold}_{low}$, the Headlight LEDs are activated.
- **Hysteresis:** A buffer zone is implemented to prevent headlight flickering during marginal light transitions.

2 Peripheral Signaling Logic

The system manages a 5-channel LED array and an acoustic horn to communicate vehicle status.

- **Directional Indicators:** PWM-driven blinking (approx. 2Hz) for Right/Left LEDs, triggered by the steering angle δ in the trajectory planner.
- **Drive Indicators:**
 - **Reverse:** Rear LEDs activate when the FSM selects a negative velocity primitive.
 - **Brake:** High-intensity rear LED activation during deceleration phases.
- **Headlights:** Controlled by the LDR-ADC logic, independent of the parking state.

3 Finite State Machine (FSM) Transitions

The parking logic operates as a deterministic FSM, with peripheral triggers tied to state transitions.

State	Input Source	Peripheral Output	Transition Condition
IDLE	CAN/UART Command	Headlights (LDR)	Start Bit Received
SEARCH	US-100 + Encoders	Blinking LEDs	Spot detected ($L_{gap} > L_{veh}$)
PLAN	CPU Math Kernel	All Stop	Path Γ generated
EXECUTE	Encoder PID	Rev/Blink LEDs	Pose Error $< \epsilon$
BLOCKED	US-100 ($< 10\text{cm}$)	Brake + Horn (10s/5s)	Path obstructed
FINISH	Encoder Feedback	Brake LEDs On	Final orientation reached

4 Conclusion

The integration of high-resolution magnetic encoders for mapping and US-100/LDR sensors for environmental interaction provides a robust framework for the Bosch Future Mobility Challenge. The non-blocking execution of signaling (LEDs/Horn) ensures the vehicle remains responsive while executing complex kinematic trajectories.